

# Marco Bustaffa

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Software developer and Data Science Master's candidate specializing in Big Data optimization, LLM systems, and reinforcement learning. Combines hands-on experience architecting scalable data pipelines and distributed systems with applied AI engineering, delivering measurable gains in storage efficiency, query performance, and system reliability. Skilled across the full ML lifecycle, from RAG pipelines and model evaluation to production-grade backend engineering.

## Experience

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**AI Engineer (Consultant)**, One-Sky-Solutions Mar 2026 – Jun 2026

- **Legal AI Pipeline (PoC):** Engineered an end-to-end Proof of Concept for automated legal document parsing, information extraction, and data validation using Large Language Models.
- **LLM Evaluation & Observability:** Established a robust evaluation and observability framework using Langfuse, achieving a 92% accuracy rate in data extraction and legal proceeding classification.

**Research Intern**, UNSW AI Institute Sep 2025 – Feb 2026

- **CRUISE Research Group:** Conducted research on multi-agent reinforcement learning for my MSc Thesis ([cruiseresearchgroup.github.io](https://cruiseresearchgroup.github.io)).

**Software Developer**, Tec Systems Engineering S.r.l June 2022 – October 2023

- **Data Infrastructure & Optimization:** Partnered with the Head of R&D to architect a Big Data storage solution using Apache Hadoop (HDFS) and Parquet; achieved a 70% reduction in storage overhead and a 30% acceleration in analytical query speeds.
- **Systems Architecture:** Led the transition from monolithic systems to a Cloud Native, microservices-oriented architecture. Engineered a roadmap that projected a 20% reduction in maintenance costs and a 30% boost in both application availability and scalability.
- **Data Engineering:** Developed a backend module in Java for the real-time exchange of vehicular traffic data with Enel X, optimizing cross-system data fidelity and communication.
- **Core Software Engineering:** Maintained mission-critical Java backend components and optimized OpenVPN network protocols to ensure secure, high-availability communication across distributed systems.

## Education

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**University of Padua**, Bachelor's Degree in Computer Science Sep 2019 – Dec 2022

**University of Padua**, Master's Degree in Data Science Sep 2023 – Jul 2026

## Projects

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**AnyLight: Multi-Agent RL for Traffic Control** UNSW AI Institute

- **Deep Reinforcement Learning:** Developed a multi-agent reinforcement learning system using PyTorch and PPO to coordinate entire traffic-signal networks.
- **Generalization:** Designed the architecture to generalize across intersections of any shape, significantly improving urban traffic management (MSc thesis).

**Env-SoundNet: Bio-inspired Sound Classification** [github.com/env-soundnet](https://github.com/env-soundnet)

- **Deep Learning Pipeline:** Engineered a neural pipeline for real-time environmental sound classification using Spiking Neural Networks (SNNs), targeting low-power neuromorphic hardware applications.
- **Signal Processing & Encoding:** Utilized Librosa and TorchAudio to process raw audio into Mel-spectrograms; implemented bio-inspired spike encoding techniques including **Delta Modulation** and **Step Forward** encoding.

- **Model Architecture:** Developed and trained Convolutional SNNs (C-SNN) using **PyTorch** and **Snntorch**, achieving high accuracy on the ESC-10 dataset while significantly reducing computational FLOPs compared to traditional CNNs.
- **Efficiency Analysis:** Conducted rigorous performance profiling to evaluate the trade-offs between temporal resolution, latency, and energy consumption, demonstrating the potential for sustainable "AI on the Edge" solutions.

#### Login Warrior: Security Anomaly Detection

[github.com/Club-Swendwich](https://github.com/Club-Swendwich)

- **Data Analytics:** Partnered with **Zucchetti** to architect a software solution for identifying suspicious activity within massive datasets of user login records.
- **High-Dimensional Visualization:** Developed a high-performance visualization engine using **React** and **WebGL** to plot large-scale security events, enabling intuitive pattern recognition for anomaly detection.
- **Engineering Leadership:** Managed the full SDLC within a team environment, focusing on scalable backend integration and rigorous documentation to solve real-world cybersecurity challenges.

### Digital Skills

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**Machine Learning & AI:** Deep/Multi-Agent RL, LLMs & RAG, LLM Evals (Langfuse), PyTorch, Spiking Neural Networks, Scikit-learn.

**Data Science & Analytics:** Python, R, SQL, Pandas, NumPy, Matplotlib.

**Big Data & Infrastructure:** Apache Hadoop, Apache Spark, Docker, NoSQL (MongoDB, InfluxDB), Linux.

**Software Engineering:** Java, C++, JavaScript, Git, React, Angular, REST APIs, Cloud-Native.